

Task 1: Investigate inode structures using commands such as `ls -li`, `stat`, and `find`. Create hard links and symbolic links and explain their impact on inode allocation.

```
shreyas@Grigori:~$ mkdir inode_test
shreyas@Grigori:~$ cd inode_test
shreyas@Grigori:~/inode_test$ echo "This is the original file." > original.txt
shreyas@Grigori:~/inode_test$ ls -li original.txt
13492 original.txt
shreyas@Grigori:~/inode_test$ stat original.txt
  File: original.txt
  Size: 27          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 13492         Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/ shreyas)   Gid: ( 1000/ shreyas)
Access: 2026-09-27 07:00:17.856063903 +0000
Modify: 2026-09-27 07:00:17.856063903 +0000
Change: 2026-09-27 07:00:17.856063903 +0000
Birth: 2026-09-27 07:00:17.856063903 +0000
shreyas@Grigori:~/inode_test$ ln original.txt hardlink.txt
shreyas@Grigori:~/inode_test$ ls -li original.txt hardlink.txt
13492 -rw-r--r--  2 shreyas shreyas 27 Sep 27 07:00 hardlink.txt
13492 -rw-r--r--  2 shreyas shreyas 27 Sep 27 07:00 original.txt
shreyas@Grigori:~/inode_test$ ln -s original.txt symlink.txt
shreyas@Grigori:~/inode_test$ ls -li original.txt hardlink.txt symlink.txt
13492 -rw-r--r--  2 shreyas shreyas 27 Sep 27 07:00 hardlink.txt
13492 -rw-r--r--  2 shreyas shreyas 27 Sep 27 07:00 original.txt
12209 lrwxrwxrwx  1 shreyas shreyas 12 Sep 27 07:00 symlink.txt -> original.txt
shreyas@Grigori:~/inode_test$ readlink symlink.txt
original.txt
shreyas@Grigori:~/inode_test$ stat original.txt
stat hardlink.txt
stat symlink.txt
  File: original.txt
  Size: 27          Blocks: 8          IO Block: 4096   regular file
  File: original.txt
  Size: 27          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 13492         Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/ shreyas)   Gid: ( 1000/ shreyas)
Access: 2026-09-27 07:00:17.856063903 +0000
Modify: 2026-09-27 07:00:17.856063903 +0000
Change: 2026-09-27 07:00:36.109584054 +0000
Birth: 2026-09-27 07:00:17.856063903 +0000
  File: hardlink.txt
  Size: 27          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 13492         Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/ shreyas)   Gid: ( 1000/ shreyas)
Access: 2026-09-27 07:00:17.856063903 +0000
Modify: 2026-09-27 07:00:17.856063903 +0000
Change: 2026-09-27 07:00:36.109584054 +0000
Birth: 2026-09-27 07:00:17.856063903 +0000
  File: symlink.txt -> original.txt
  Size: 12          Blocks: 0          IO Block: 4096   symbolic link
Device: 8,48      Inode: 12209         Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/ shreyas)   Gid: ( 1000/ shreyas)
Access: 2026-09-27 07:00:51.834300564 +0000
Modify: 2026-09-27 07:00:44.600730907 +0000
Change: 2026-09-27 07:00:44.600730907 +0000
Birth: 2026-09-27 07:00:44.600730907 +0000
```

```

shreyas@Grigori:~/inode_test$ find . -inum 13492
./hardlink.txt
./original.txt
shreyas@Grigori:~/inode_test$ echo "Modified through hard link." >> hardlink.txt
shreyas@Grigori:~/inode_test$ cat original.txt
This is the original file.
Modified through hard link.
shreyas@Grigori:~/inode_test$ echo "Modified through symbolic link." >> symlink.txt
shreyas@Grigori:~/inode_test$ cat original.txt
This is the original file.
Modified through hard link.
Modified through symbolic link.
shreyas@Grigori:~/inode_test$ stat original.txt
  File: original.txt
  Size: 87          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 13492        Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/ shreyas)   Gid: ( 1000/ shreyas)
Access: 2026-09-27 07:02:21.846014688 +0000
Modify: 2026-09-27 07:02:16.328590177 +0000
Change: 2026-09-27 07:02:16.328590177 +0000
 Birth: 2026-09-27 07:00:17.856063903 +0000
shreyas@Grigori:~/inode_test$ rm original.txt
shreyas@Grigori:~/inode_test$ cat hardlink.txt
This is the original file.
Modified through hard link.
Modified through symbolic link.
shreyas@Grigori:~/inode_test$ cat symlink.txt
cat: symlink.txt: No such file or directory
shreyas@Grigori:~/inode_test$ echo "This is the original file." > original.txt

shreyas@Grigori:~/inode_test$ ls -li
total 8
13492 -rw-r--r-- 1 shreyas shreyas 87 Sep 27 07:02 hardlink.txt
 1030 -rw-r--r-- 1 shreyas shreyas 27 Sep 27 07:14 original.txt
12209 lrwxrwxrwx 1 shreyas shreyas 12 Sep 27 07:00 symlink.txt -> original.txt
shreyas@Grigori:~/inode_test$ cat symlink.txt
This is the original file.
shreyas@Grigori:~/inode_test$ stat original.txt
  File: original.txt
  Size: 27          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 1030        Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/ shreyas)   Gid: ( 1000/ shreyas)
Access: 2026-09-27 07:15:10.747930549 +0000
Modify: 2026-09-27 07:14:43.758247388 +0000
Change: 2026-09-27 07:14:43.758247388 +0000
 Birth: 2026-09-27 07:14:43.758247388 +0000
shreyas@Grigori:~/inode_test$ stat hardlink.txt
  File: hardlink.txt
  Size: 87          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 13492        Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/ shreyas)   Gid: ( 1000/ shreyas)
Access: 2026-09-27 07:02:35.863696058 +0000
Modify: 2026-09-27 07:02:16.328590177 +0000
Change: 2026-09-27 07:02:31.438212218 +0000
 Birth: 2026-09-27 07:00:17.856063903 +0000
shreyas@Grigori:~/inode_test$ stat symlink.txt
  File: symlink.txt -> original.txt
  Size: 12          Blocks: 0          IO Block: 4096   symbolic link
Device: 8,48      Inode: 12209        Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/ shreyas)   Gid: ( 1000/ shreyas)
Change: 2026-09-27 07:00:44.600730907 +0000
 Birth: 2026-09-27 07:00:44.600730907 +0000
shreyas@Grigori:~/inode_test$ readlink symlink.txt
original.txt
shreyas@Grigori:~/inode_test$ cd ..
shreyas@Grigori:~$ rm -rf inode_test

```

Task 2: Design a program using mmap() to perform file reading and writing. Compare memory-mapped I/O with traditional read/write operations in terms of efficiency and implementation complexity.

```
shreyas@Grigori:~$ nano mmap_file.c
shreyas@Grigori:~$ gcc mmap_file.c -o mmap_file
shreyas@Grigori:~$ ./mmap_file
Original file content:
This file is being accessed using memory mapping.

Modified file content:
Modified using mmap().
ssed using memory mapping.
shreyas@Grigori:~$ cat mmap_input.txt
Modified using mmap().
ssed using memory mapping.
shreyas@Grigori:~$ ls -l mmap_input.txt
-rw-r--r-- 1 shreyas shreyas 50 Sep 27 07:17 mmap_input.txt
shreyas@Grigori:~$ nano traditional_io.c
shreyas@Grigori:~$ gcc traditional_io.c -o traditional_io
shreyas@Grigori:~$ ./traditional_io
File content using read():
Modified using mmap().
ssed using memory mapping.
shreyas@Grigori:~$ ./mmap_file
Original file content:
This file is being accessed using memory mapping.

Modified file content:
Modified using mmap().
ssed using memory mapping.
shreyas@Grigori:~$ ./traditional_io
File content using read():
Modified using mmap().
```

```

shreyas@grigori:~$ cat mmap_input.txt
Modified using mmap().
ssed using memory mapping.
shreyas@grigori:~$ ls -l mmap_input.txt
-rw-r--r-- 1 shreyas shreyas 50 Sep 27 07:17 mmap_input.txt
shreyas@grigori:~$ nano traditional_io.c
shreyas@grigori:~$ gcc traditional_io.c -o traditional_io
shreyas@grigori:~$ ./traditional_io
File content using read():
Modified using mmap().
ssed using memory mapping.
shreyas@grigori:~$ ./mmap_file
Original file content:
This file is being accessed using memory mapping.

Modified file content:
Modified using mmap().
ssed using memory mapping.
shreyas@grigori:~$ ./traditional_io
File content using read():
Modified using mmap().
ssed using memory mapping.
shreyas@grigori:~$ rm mmap_file traditional_io mmap_file.c traditional_io.c mmap_input.txt
shreyas@grigori:~$ nano 7.2
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

#define BUFFER_SIZE 1024

int main()
{
    int fd;
    char buffer[BUFFER_SIZE];
    ssize_t bytes;

    fd = open("mmap_input.txt", O_RDONLY);

    if (fd < 0)
    {
        perror("open");
        return 1;
    }

    bytes = read(fd, buffer, BUFFER_SIZE - 1);

    if (bytes < 0)
    {
        perror("read");
        close(fd);
    }
}

```